

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

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COURSE CODE: CSA0708

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COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation*

tools Assessment Tool Weightage: 20%

QUESTIONS: 40 Total Marks:40 Annexure A

Mock Interview

QUESTIONS:40 Total Marks:40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)
27. How would you present your simulation results to a client? (BL4)(1)

28. Explain jitter using a real-world analogy (BL5)(1)
29. Describe how you tested network performance in a lab (BL4)(1)
30. Explain the importance of performance metrics in one structured answer (BL5)(1)
31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
32. Why are you interested in networking and simulation tools? (BL5)(1)
33. What makes you confident in analyzing network performance? (BL5)(1)
34. Explain a concept you are very strong in (related to networking) (BL5)(1)
35. How would you handle a question you don't know? (BL5)(1)
36. Describe a situation where you solved a technical problem (BL4)(1)
37. Why should we select you for a networking role? (BL5)(1)
38. What challenges did you face while learning Packet Tracer? (BL5)(1)
39. How do you improve your technical skills regularly? (BL5)(1)
40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION					
Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANSWERS

1. Analyze how network congestion affects packet delay and packet loss.
- A) Network congestion increases packet delay because packets wait in queues, and packet loss occurs when network buffers become full.

2. Compare the impact of star and mesh topologies on network reliability and fault tolerance.

A) Star topology depends on a central device, while mesh topology provides multiple paths and therefore offers better reliability and fault tolerance.

3. Why does increasing the number of routers between source and destination increase end-to-end delay?

A) Each additional router adds processing, queuing, and transmission delay, which increases the total end-to-end delay.

4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic.

A) QoS prioritizes voice and video packets, reducing delay, jitter, and packet loss for smoother communication.

5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication.

A) Fiber optics provide higher speed, greater bandwidth, longer transmission distance, and less electromagnetic interference than copper cables.

6. Analyze how network segmentation using VLANs improves performance and security.

A) VLANs reduce broadcast traffic and isolate different user groups, improving both network performance and security.

7. Compare the effects of TCP and UDP on network performance for different applications.

A) TCP provides reliable delivery but has more overhead, while UDP is faster and suitable for real-time applications such as gaming and video calls.

8. Why is load balancing important in large enterprise networks?

A) Load balancing distributes traffic across multiple links or servers, improving performance, availability, and reliability.

9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization.

A) Frequent routing updates improve route accuracy but consume additional bandwidth and processing resources.

10. Evaluate the trade-off between network security measures and overall network performance.

A) Security measures improve protection but may add processing overhead, causing a slight reduction in network performance.

11. A network has high delay—how will you identify and fix the issue?

A) I would use ping and traceroute to identify the problem and fix congestion, routing issues, or insufficient bandwidth.

12. If throughput is low despite high bandwidth, what could be the reasons?

A) Low throughput may result from congestion, packet loss, high latency, poor configuration, or device limitations.

13. How would you reduce packet loss in a congested network?

A) Packet loss can be reduced by increasing bandwidth, applying QoS, managing traffic, and upgrading overloaded devices.

14. A VoIP call is breaking frequently—how would you troubleshoot?

A) I would check bandwidth, delay, jitter, packet loss, and QoS configuration to identify the cause.

15. Given a network with high jitter, suggest improvements.

A) High jitter can be reduced by using QoS, reducing congestion, increasing bandwidth, and using jitter buffers.

16. How will you simulate a congested network in Packet Tracer?

A) I would create heavy traffic between devices over a limited-bandwidth link and observe delay and packet loss.

17. If packets are delayed in a router, what steps will you take?

A) I would check router CPU usage, interface bandwidth, queue congestion, and routing configuration.

18. How would you optimize a network for online gaming?

A) I would reduce latency, jitter, and packet loss using QoS, high-speed links, and efficient routing.

19. Design a small network and explain how you will measure its performance.

A) I would connect PCs through switches and routers and measure performance using throughput, delay, packet loss, and ping tests.

20. If RTT is very high, how will you diagnose the issue?

A) I would use ping and traceroute to identify congestion, routing problems, long paths, or slow network links.

21. Explain network performance to a non-technical person.

A) Network performance means how fast, smoothly, and reliably information travels between devices.

22. Describe throughput using a real-life example.

A) Throughput is like the number of vehicles that successfully pass through a road in a specific amount of time.

23. Present how packet loss affects video streaming.

A) Packet loss can cause buffering, poor video quality, frozen frames, and interruptions during streaming.

24. Explain delay types clearly in under 1 minute.

A) The four main delay types are processing delay, queuing delay, transmission delay, and propagation delay.

25. Describe Cisco Packet Tracer in a professional way.

A) Cisco Packet Tracer is a network simulation tool used to design, configure, test, and troubleshoot virtual network environments.

26. Explain QoS in simple and technical terms.

A) QoS is a network mechanism that prioritizes important traffic to reduce delay, jitter, and packet loss.

27. How would you present your simulation results to a client?

A) I would present the network diagram, test results, performance metrics, graphs, problems identified, and recommended solutions.

28. Explain jitter using a real-world analogy.

A) Jitter is like buses arriving at irregular times instead of arriving according to a fixed schedule.

29. Describe how you tested network performance in a lab.

A) I tested network performance by sending packets and measuring throughput, delay, jitter, and packet loss.

30. Explain the importance of performance metrics in one structured answer.

A) Performance metrics such as throughput, delay, jitter, and packet loss help measure network speed, quality, reliability, and efficiency.

31. Introduce yourself and explain your knowledge of network performance.

A) I am a networking learner with knowledge of throughput, delay, jitter, packet loss, and practical experience using Cisco Packet Tracer.

32. Why are you interested in networking and simulation tools?

A) I am interested because networking and simulation tools help me practically design, test, analyze, and troubleshoot networks.

33. What makes you confident in analyzing network performance?

A) My understanding of performance metrics and practical experience with network simulation tools make me confident.

34. Explain a concept you are very strong in related to networking.

A) I am strong in routing, where routers analyze available paths and select the best route for data transmission.

35. How would you handle a question you don't know?

A) I would remain calm, analyze the question, explain what I know, and learn the correct answer afterward.

36. Describe a situation where you solved a technical problem.

A) I identified a network connectivity problem, checked the configuration step by step, corrected the issue, and verified the connection.

37. Why should we select you for a networking role?

A) You should select me because I have networking fundamentals, practical simulation knowledge, problem-solving skills, and a willingness to learn.

38. What challenges did you face while learning Packet Tracer?

A) I initially faced configuration and connectivity issues, but I improved through regular practice and troubleshooting.

39. How do you improve your technical skills regularly?

A) I improve my skills by practicing networking labs, learning new concepts, and working with

simulation tools regularly.

40. Demonstrate confidence while explaining a complex concept simply.

A) I explain complex concepts by breaking them into simple steps and using practical examples so that anyone can understand them.